YAKEEN NEET 2.0

2026

The p-Block Elements

Inorganic Chemistry | Lecture - 03

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Topics to be covered

1. Oxidation States & Chemical Reactivity (13th & 14th group) (group wise)
2.

Oxidation States and Chemical Reactivity

NEET syllabus of p Block

UNIT 10: P- BLOCK ELEMENTS

Group -13 to Group 18 Elements

General Introduction: Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour of the first element in each group.

(ii) Reactivity towards acids and alkalies

w/i water insoluble
w/s water soluble

Elemental Boron → Acid & Base ~~Reaction~~

Al → Reaction ✓✓

Amphoteric Nature → Acid & Base.

(A) $Al + HCl \longrightarrow Al^{3+}_{(aq)} + Cl^-_{(aq)} + H_2\uparrow$

$(AlCl_3)$

Al → Al^{3+}, e^-

HCl → Acid → H^+ → $H_2\uparrow$

(B) $Al_{(s)} + NaOH_{(aq)} \rightarrow Na[Al(OH)_4] + H_2\uparrow$

$NaOH_{(aq)}$ → $(NaOH + H_2O)$

$Na[Al(OH)_4]$ w/s

Sod. tetrahydroxidoaluminate(III)

This Rxn is used to Remove blockage From drainage Pipes

$Al + C.HNO_3 \longrightarrow$ Passivation

Formation of Protective oxide layer on metal surface.

Be, Al, Fe, Cr ; Actinoids

$\xrightarrow{C.HNO_3}$ (Passivation)

$$Al + H_2O + NaOH \longrightarrow Al(OH)_3\downarrow \longrightarrow Na^+[Al(OH)_4]^- + H_2\uparrow$$

$Al \rightarrow Al^{3+} + e^-$

$H_2O \rightarrow H^+ + OH^-$

$NaOH \rightarrow Na^+ + OH^-$

H_2

$Al(OH)_3\downarrow$ white gel. ppt

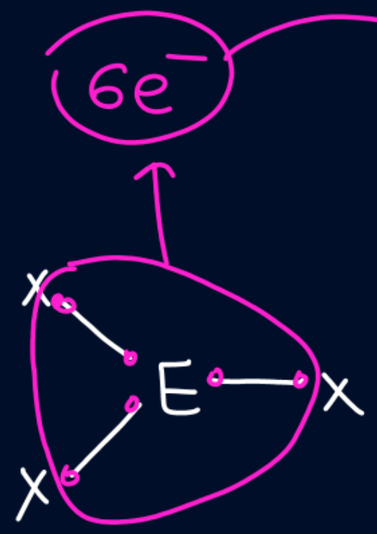
$Na[Al(OH)_4]$

(iii) Reactivity towards halogens

$E + X_2 \longrightarrow \boxed{EX_3 \text{ trihalides.}}$

Except TlI_3

$\cancel{Tl^{3+}}$ $\cancel{3I^-}$



(Hypovalent)

- e^- deficient in nature (Lewis Acid)
- Can react with e^- Rich species

Ex

$BCl_3 + \ddot{N}H_3 \longrightarrow H_3N: \rightarrow BCl_3$ → e^- deficiency Removed

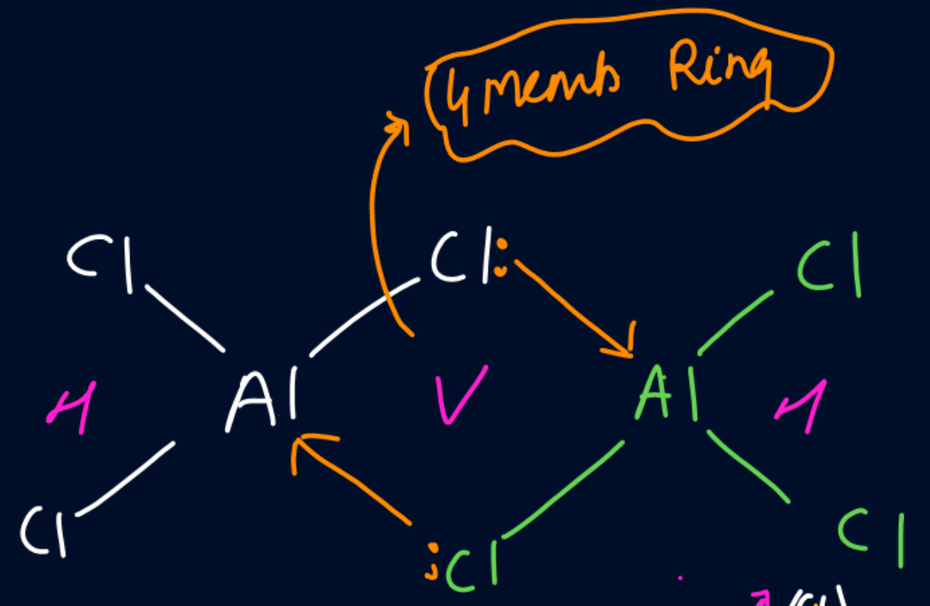
L.A L.B

$AlCl_3 + NH_3 \longrightarrow H_3N: \rightarrow AlCl_3$

NOTE

$AlCl_3$ Undergoes dimerisation to remove its e^- deficiency.

$AlCl_3$ e^- deficient

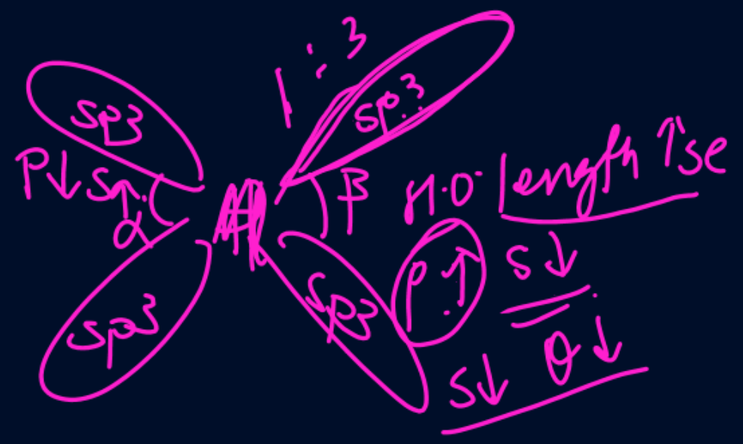


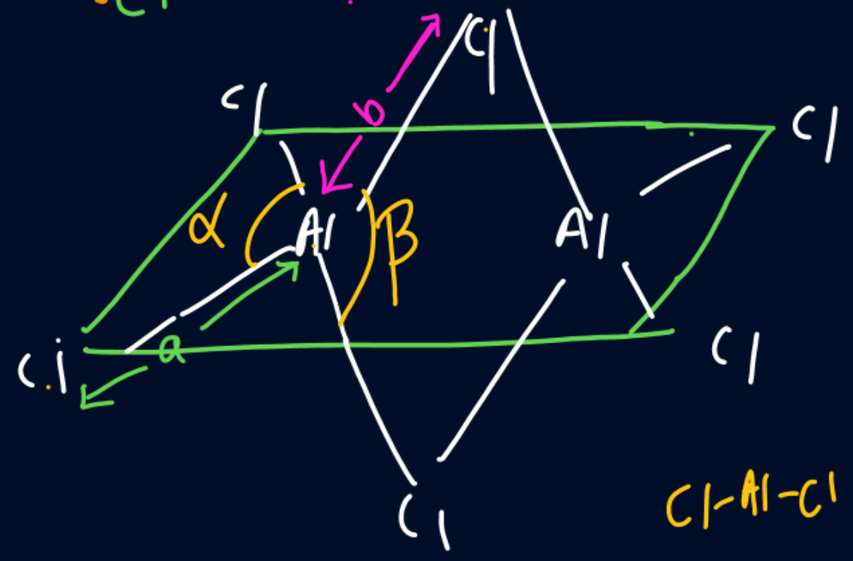
→ Halogen bridged dimer

→ e^- deficiency Removed.

→ Al → sp^3, tetrahedral.

→ two tetrahedral Unites





$a < b$

$\alpha > \beta$

Al–Cl Terminal B.L < Al–Cl Bridging B.L

Cl–Al–Cl terminal θ > Cl–Al–Cl Bridging angle

$BCl_3 + BCl_3 \longrightarrow \cancel{B_2Cl_6}$

~~dimerisation~~

$\because$ due to small size of Boron.

~~B_2F_6~~

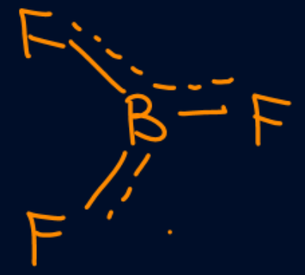
$BF_3 > BCl_3 > BBr_3 > BI_3$

(B·B Extent)

strength

effective B·B

No need to dimerise



$AlF_3 + AlF_3 \longrightarrow \cancel{Al_2F_6}$

AlF_3 ~~Covalent~~ Ionic ✓

Al^{3+} { $3F^-$

octet complete

Octet complete

V.Imp

Problem 11.2

White fumes appear around the bottle of anhydrous aluminium chloride. Give reason.

Solution

Anhydrous aluminium chloride is partially hydrolysed with atmospheric moisture to liberate HCl gas. Moist HCl appears white in colour.

$$AlCl_{3(s)} + H_2O \longrightarrow Al(OH)_3 + HCl\uparrow$$

(moisture)

Al^{3+} Cl^- H^+ OH^-

white fumes

Oxidation state & trends in Chemical reactivity : Group 14th

$ns^2 np^2$

- Common oxidation states → +4 and +2.

- Carbon also exhibits negative oxidation states. (Carbides)

- Compounds in +4 oxidation state are generally covalent in nature.

high

a+b+c+d

Reason : Sum of the first four ionization enthalpies is very high.

$E \xrightarrow{a} E^{+} \xrightarrow{b} E^{2+} \xrightarrow{c} E^{3+} \xrightarrow{d} E^{4+}$

- In heavier members the tendency to show +2 oxidation state increases in the sequence Ge < Sn < Pb.

Reason :

* Inert pair effect

- Lead compounds in +2 state are stable and in +4 state are strong oxidising agents.

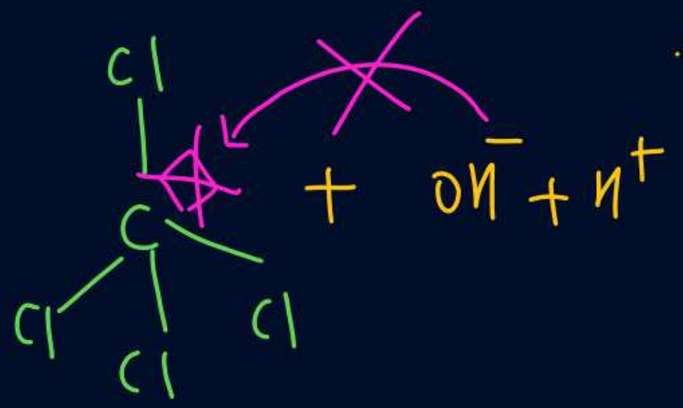
Reason : Inert pair effect

$CF_4 + 2F^- \rightarrow [CF_6]^{2-}$ $SiF_4 + 2F^- \rightarrow [SiF_6]^{2-}$

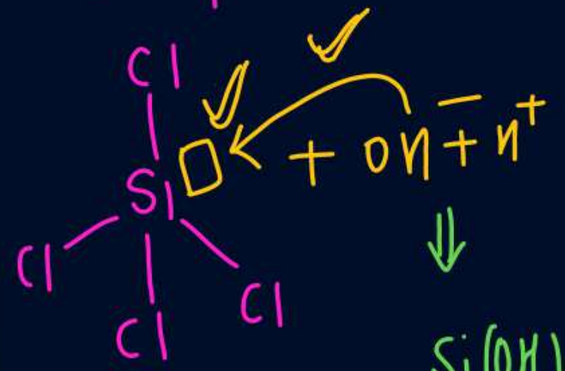
- Due to presence of d orbital in heavier members their halides undergo hydrolysis and have tendency to form complexes by accepting electron pairs from donor species. For example, the species like, SiF_6^{2-}, $[GeCl_6]^{2-}$, $[Sn(OH)_6]^{2-}$ exist where the hybridisation of the central atom is sp^3d^2.

$CCl_4 \rightarrow$ Hydrolysis

Reason due to absence of vacant orbital at C.A (carbon)



$SiCl_4$



$Si(OH)_4 + HCl$

Final Products

Hydrolysis

Addition of H_2O

But H_2O Break Hoga < H^+, OH^-

Coordinate Bond

$A + OH^-$

Req. ⇓

→ Vacant orbital

→ +δ charge

$\underset{+\delta}{A} - L + OH^- + H^+ \rightarrow A\,OH + HL$

Ionic Comp. tendency.

Hydration

Addition of H_2O Bina toote

ion-dipole & dipole-dipole

(i) Reactivity towards oxygen

$E + O_2 \longrightarrow EO / EO_2$

oxides.

Ex CO
Neutral

SiO
Exist at High temp

CO_2
SiO_2

(gen)

O^{2-} of C, Si, Ge
acidic

O^{2-} Sn & Pb
Amphoteric

(ii) Reactivity towards water

$(H_2O)_{(l)}$

$C/Si/Ge/Sn/Pb + H_2O_{(l)} \longrightarrow$ No Reaction

Pb → Expected to React
But due to presence of Oxide layer on its Surface.
It does not React.

Ex

Sn Can React with

$Sn + H_2O_{(g)} \longrightarrow SnO_2 + H_2\uparrow$

(Steam)

(iii) Reactivity towards halogen

$E + X_2 \longrightarrow \overset{+2}{E}X_2 / \overset{+4}{E}X_4$

*** PbI_4 ~~exist~~

Pb^{+4} I^-
O.A R.A

[Stability of EX_2 ↑se (T→B)
— " — EX_4 ↓se (T→B)] ⇒ Inert pair effect.

EX_4 Covalent in nature (H.O.S)

NEET PYQ ← [But SnF_4 & PbF_4 are Ionic
[∵ Small size of F^-]]

$GeX_2 < \overset{+4}{Ge}X_4$

$\overset{+2}{Pb}X_2 > PbX_4$

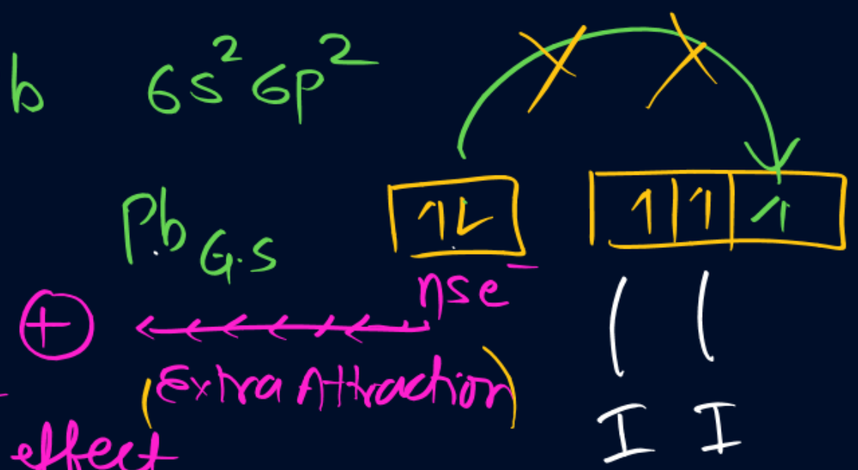
NCERT

Imp

PbI_4 does not exist.

Pb $6s^2 6p^2$

$Pb_{G.S}$



Inert pair effect

Zeff ↑sed

* due to Large B.L of initially Formed two Pb–I Bonds → less Eneregy Released which is not sufficient to Unpair 6s e^- Pair.

PbI_4 ✗ PbI_2 ✓

Hydrolysis of Group 14th halides

(i) CCl_4 → ~~Hydrolysis~~

(ii) SiF_4 → Partial Hydrolysis

(iii) $SiCl_4$ → Complete Hydrolysis

??

Ans because $[SiCl_6]^{2-}$ does not exist due steric Hindrace

$$SiCl_4 + H_2O \longrightarrow Si(OH)_4 + HCl$$

H_2O: OH^-, H^+

HCl: Cl^-, H^+

Complete Hydrolysis

$\cancel{[SiCl_6]^{2-}}$ $\cancel{\text{Exist}}$

Steric Hindrance

$$SiF_4 + H_2O \longrightarrow Si(OH)_4 + HF + [SiF_6]^{2-}$$

H_2O: H^+, OH^-

HF: H^+, F^-

Hexafluoride

Partial Hydrolysis

Problem 11. 6

$[SiF_6]^{2-}$ is known whereas $[SiCl_6]^{2-}$ not. Give possible reasons.

Solution

The main reasons are :

(i) six large chloride ions cannot be accommodated around Si^{4+} due to limitation of its size. (steric hindrance)

(ii) interaction between lone pair of chloride ion and Si^{4+} is not very strong.

$Si^{+4} \sim Cl^-$
weak
Interaction

Oxidation state & trends in Chemical reactivity : Group 15th

- Common oxidation states → –3, +3, +5.

- Tendency to show –3 decreases down the group.

Reason : To pto bottom Size↑ IE↓, Met. Ch↑ N. M. C ↓

- Stability of +5 decreases while stability of +3 increases down the group.

Reason :

Inert Pair effect

Question

The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of

[24 Jan, 2025 (Shift-I)]

1. Atomic size
2. Atomicity
3. Electron gain enthalpy
4. Electronegativity

H.W

THANK YOU